Build the Five

A paper-craft kit of the five perfect shapes, and one that fills space



DICTO · dictospheres.com · November 2026 · Nets unfolded from the solids by computer and checked: no overlaps, one glue tab on every seam.

What you need

An A4 printer, thin card or thick paper, scissors, a ruler, glue. An adult with a blunt knife or an empty ballpoint pen to score the fold lines makes the folds crisp.

The pieces, in D·I·C·T·O order

	Shape	Faces	Level
	D Dodecahedron	**12 pentagons**	**harder**
	I Icosahedron	**20 triangles**	**medium**
	C Cube	**6 squares**	**easy**
	T Tetrahedron	**4 triangles**	**easy**
	O Octahedron	**8 triangles**	**easy**
	Bonus Rhombic dodecahedron	**12 rhombi**	**harder**

Every piece has the same edge, 30 mm, so the five sit side by side at their true sizes. Each colour is one shape, and the colours match DICTO's Nest Kit.

How to build

1. Print at 100% (not "fit to page"). Measure the 50 mm bar at the bottom of each page to check.
2. Cut along the solid lines, round the grey tabs.
3. Fold every dashed line, all the same way.
4. Edges with the same number meet. Glue each grey tab under the edge carrying its number.
5. Start easy: tetrahedron, octahedron, cube. Then the icosahedron, then the dodecahedron.

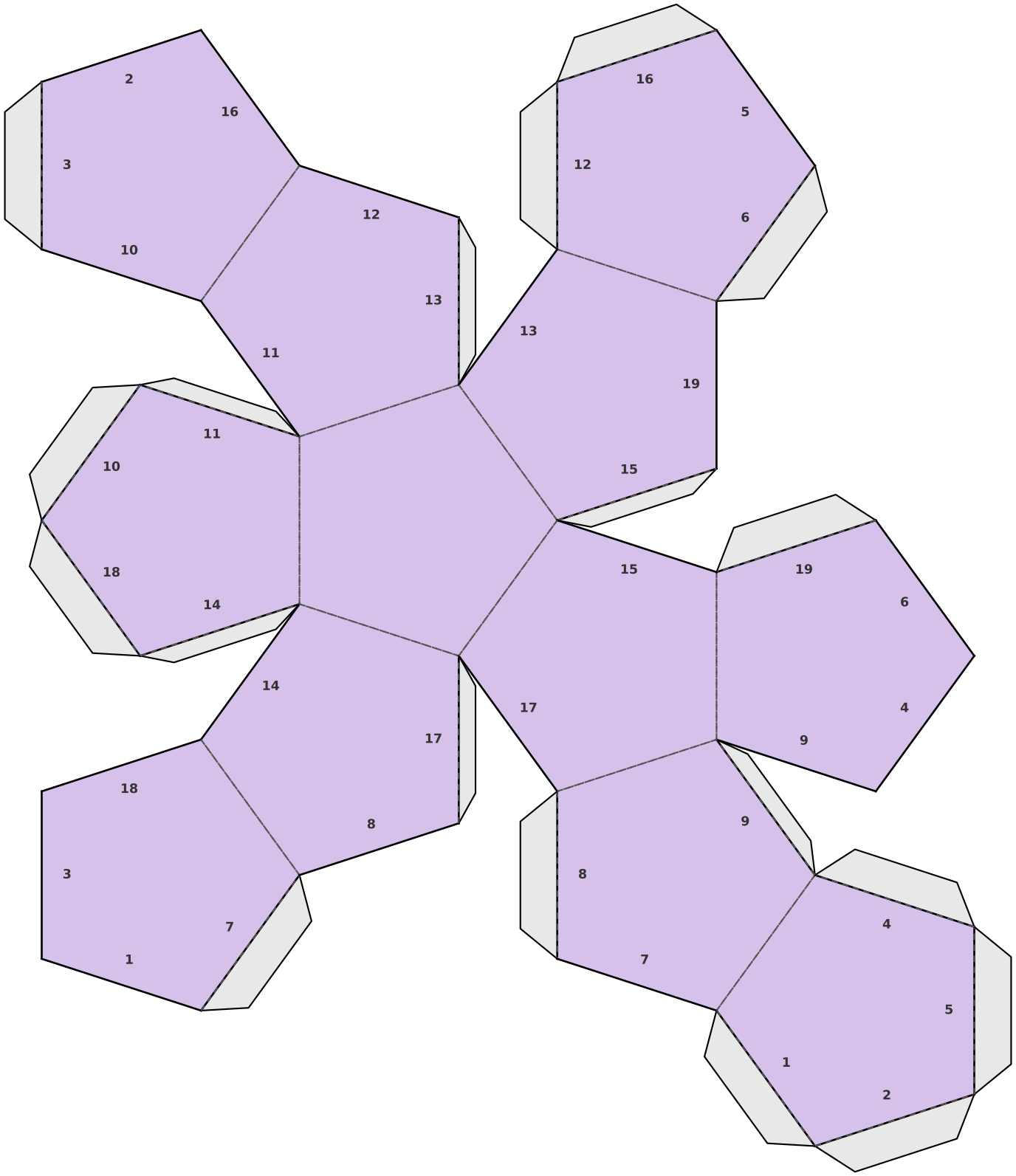
Things to try

- Count the corners, edges and faces of each: corners − edges + faces always comes to 2.
- Build several rhombic dodecahedra: they fit face to face, and enough of them fill space by themselves with no gaps, like cubes.
- Print the tetrahedron twice and the octahedron once: two tetrahedra and one octahedron fit face to face, the octet that fills space.

D · DODECAHEDRON

12 pentagons · edge 30 mm · harder

Print at 100%. Cut the solid lines, fold the dashed lines, glue the grey tabs. Edges with the same number meet.

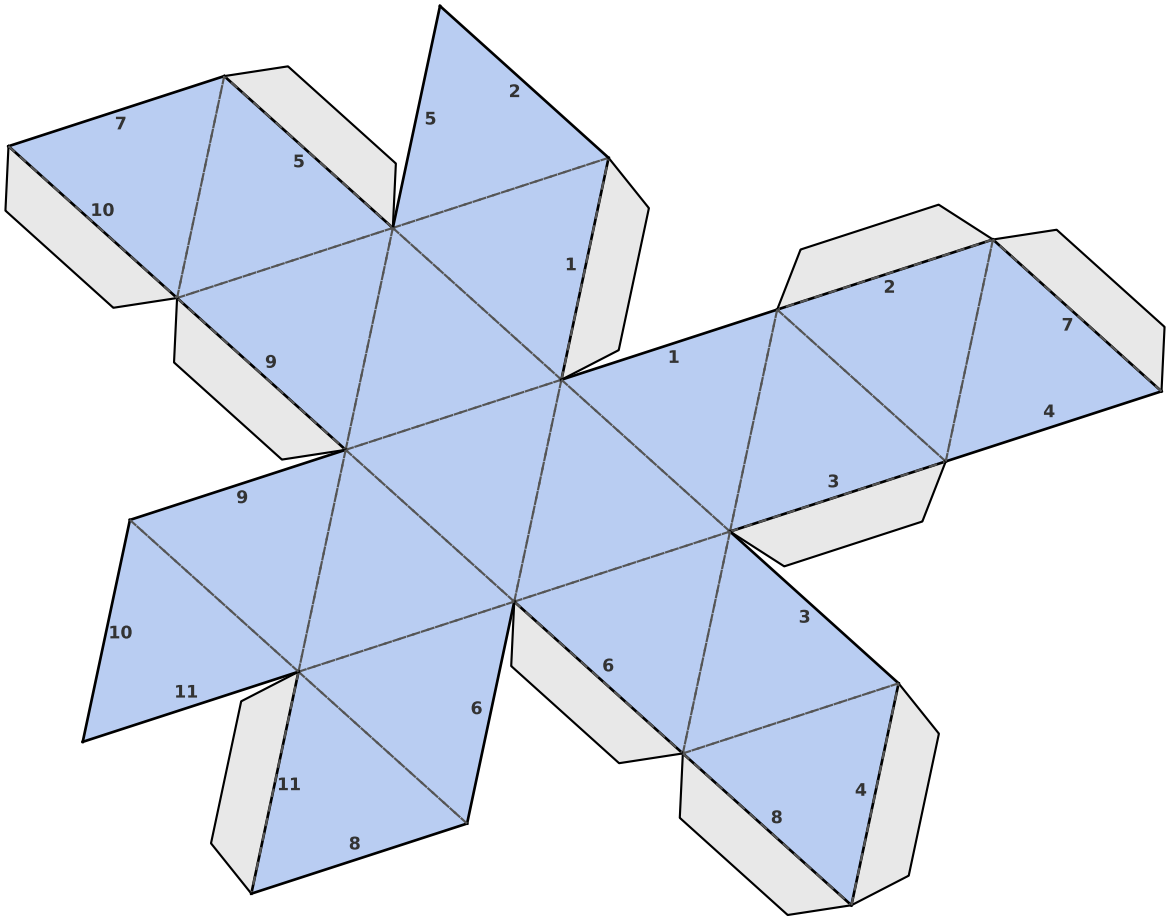


50 mm: check your print

I · ICOSAHEDRON

20 triangles · edge 30 mm · medium

Print at 100%. Cut the solid lines, fold the dashed lines, glue the grey tabs. Edges with the same number meet.

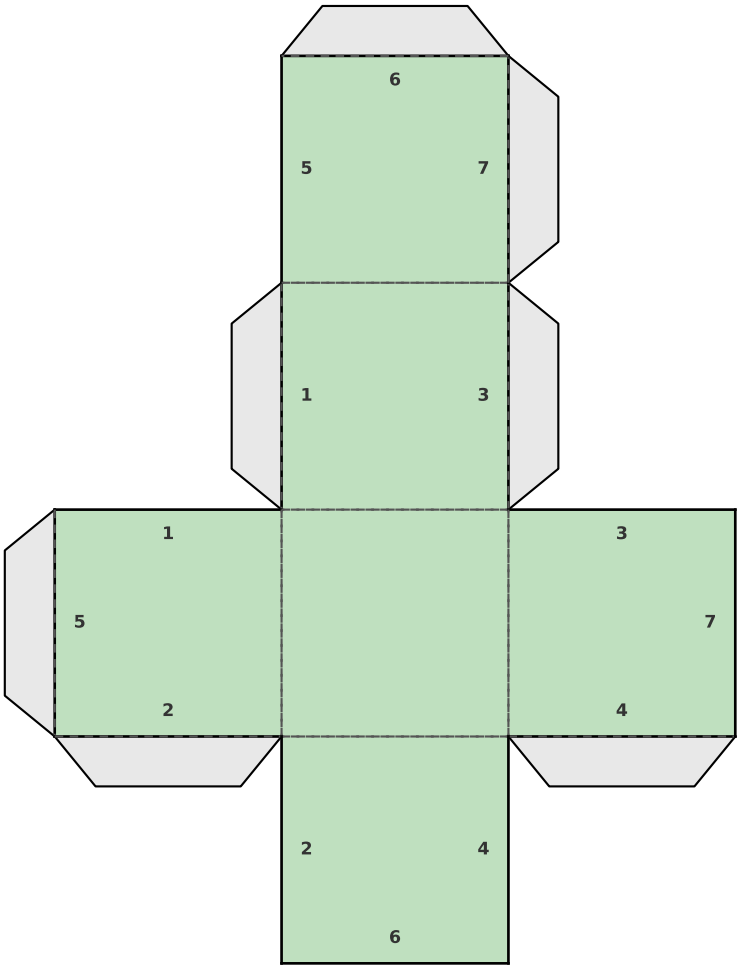


50 mm: check your print

C · CUBE

6 squares · edge 30 mm · easy

Print at 100%. Cut the solid lines, fold the dashed lines, glue the grey tabs. Edges with the same number meet.

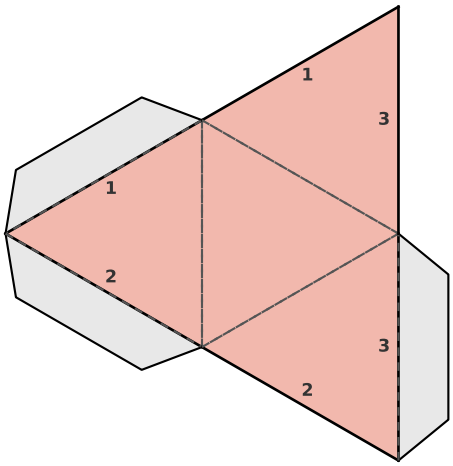


50 mm: check your print

T · TETRAHEDRON

4 triangles · edge 30 mm · easy

Print at 100%. Cut the solid lines, fold the dashed lines, glue the grey tabs. Edges with the same number meet.

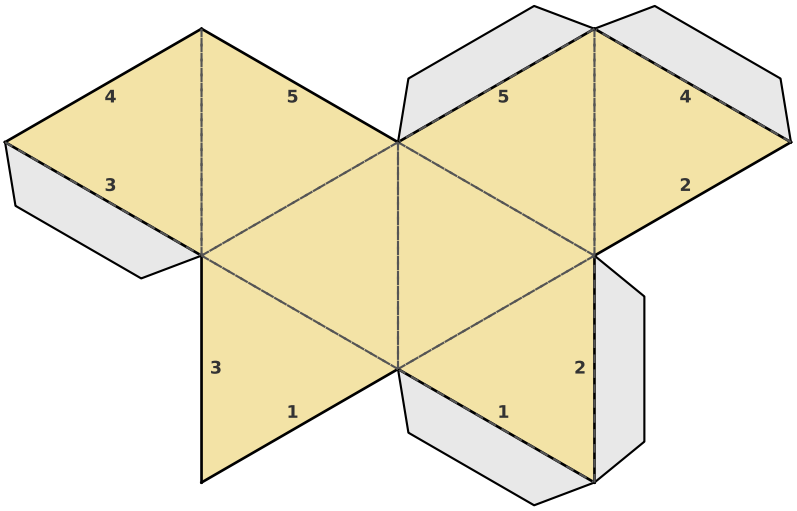


50 mm: check your print

O · OCTAHEDRON

8 triangles · edge 30 mm · easy

Print at 100%. Cut the solid lines, fold the dashed lines, glue the grey tabs. Edges with the same number meet.

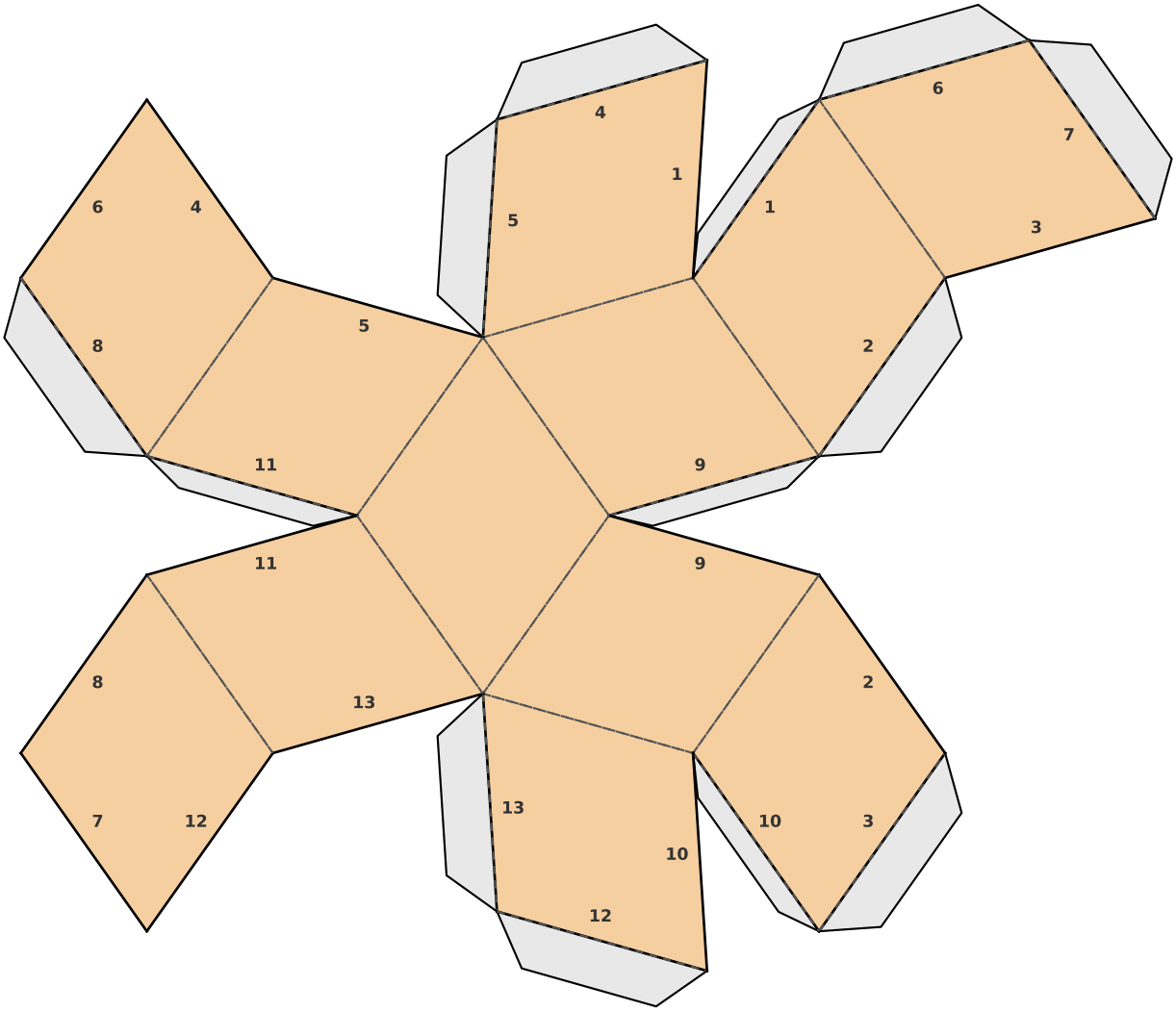


50 mm: check your print

BONUS · RHOMBIC DODECAHEDRON

12 rhombi: fills space by itself, like the cube · edge 30 mm · harder

Print at 100%. Cut the solid lines, fold the dashed lines, glue the grey tabs. Edges with the same number meet.



50 mm: check your print